

Is the energy spending of ai in a personal level even comparable to the automated industrial level of ai use?

Posted by u/mervenca • 0 points • 17 comments

I've seen a lot of posts and discussion about the ai using up a lot of energy and water, for example every prompt using up a bottle of water etc..

But to me it kind of seems so absurd to put the blame on individual use, if (i imagine) the probably huge automated systems and industrial amount of ai processing spends the energy in much higher unimaginable amounts every second..

I guess it is just a matter of ethics and everyones environmental responsibility to minimise the impact to our planet, but it really feels like futule fighting the with the windmills.

Am I wrong here? Whats your take on the problem?

Comments

u/Autobahn97 • 1 points • 2025-03-14 14:11 UTC

Bottle of water!? Sounds like some politically motivated misinformation to me. But whatever the repercussions are it really doesn't matter because its going to happen one way or another and humanity will progress, as it always has, and just need to deal with any challenges which might inadvertently occur.

u/StevenSamAI • 1 points • 2025-03-12 11:49 UTC

I think the enrgy consumption side of things is a very good point for debate.

Often raised by people on the anti-AI side of things, but IMO, one of the best arguments about a negative impact of AI. However, some of the statements made about it are crazy. We can have an extremely good idea about the amount of energy and resources used for both training and running models, and for the latter, we can usually even prove these on our own computers.

Does AI as a whole use a shit ton of energy. Yes, and I think it isn't helpful for people to brush this point aside just because they are in favour of AI.

A common response I see when people question AI energy usage is "what about netflix, what about rendering clouds, etc.". These points are not wrong, there are also other industries that do use a lot of

energy running big data centers. It's a valid part of the discussion, but it doesn't negate the point. Whataboutism doesn't really settle a valid talking point.

Your question about personal vs. industrial is an interesting one. I can download a good image generator, or even a pretty decent LLM, and run this quite well on fairly low power bit of consumer hardware. This might give me the feeling that my use of AI isn't really any different to me watching a movie, playing a game, rendering a 3D model, etc. and there is truth to that. If I'm generating images instead of rendering 3D models, then I've offset my existing usage to a different application, and there isn't a net change. If I am doing it in addition, there is an increase in energy consumed.

If hundreds of millions of people are doing this, each small amount of energy consumed adds up.

While it seems intuitive to look at a data centre vs a home computer and think the data centre is worse, as it clearly uses more energy, I think the question to ask is, "Is it more efficient?" and "What is the energy source?".

There are a lot of efficiency gains of doing things at scale. I don't know as much about image generators, but with LLM's, a server can process lots of prompts in parallel. e.g. one prompt might get a response at a speed of 100 tokens per second, but the same hardware can process 10 prompts in parallel, achieving a total speed of 900 tokens per second. These efficiency gains reduce the energy per token, so the economy of scale kicks in from an energy perspective. Sure, there is the overhead in routing it around the internet, and it gets complex to figure out. However these are important considerations.

Similarly, if my local energy provider is burning fuel to pipe electricity to my PC, and a data centre has a nuclear power plant + wind farms providing 90% of its energy, then it is likely the case that for the overall use of millions of AI users, the data centre is the more environmentally friendly option.

All I'm getting at is that there is more nuance to these points, and they are worth of discussion. However it is likely too complex to pin down, and will vary between each individual user and each data centre.

Regarding the training costs in energy. They can be very high, but they are then spread out over all of the inference uses, and a gain, the source of the energy becomes relevant.

tbc...

u/StevenSamAI • 1 points • 2025-03-12 11:49 UTC

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I mentioned whataboutism not being really addressing the argument made about AI, and I think that's true, there is no need to divert the argument and avoid the question. However, if people are using energy consumption of AI as a reason to tell people they shouldn't use it, and that same person is using equal or more amounts of GPU/data centre energy for other things, then it is hypocritical.

A final point to consider, is the efficiencies AI will, and already does offer that reduce energy consumption. Among other things, I am a software developer. If I'm working on a software feature, I used to google around a lot, find tutorials, read documentation, slowly type code one line at a time, and when it went wrong I'd search the error, read stack overflow, post questions, and slowly try to piece together a solution. Realistically, this took a lot of energy from my PC, the internet

connections, and multiple servers and data centers. Compare this to my current programming method which is me using Claude Sonnet LLM for almost everything. On average, I'd say that I can create a typical feature 5-10 times faster than I could previously. So for that aspect of my work, I might only need my computer on for 1 day instead of 5. That is a noticeable efficiency improvement. I think there will be a lot more examples of things like this as well.

Also, while it is an example that some artists might not like to hear, I do use AI to create images that I would previously have used an artist to create. Simple commercial stuff like banner images for a website, etc. I don't need a work of art, or a masterpiece, or care about the soul or the process. I just want a nice banner image that is on brand for my website, and often just prompting is fit for purpose for my needs. So, instead of the lengthy process I used to take with artists, this is a tiny bit of prompting, and maybe a crop and an upscale, or outpainting. I think the energy I use now to create such things is comparable to the energy used to convey the requirements to an artist, without even considering their process of making the image.

u/victorc25 • 1 points • 2025-03-12 07:13 UTC

Humans produce more carbon emissions than the

AI <https://aiadvisoryboards.wordpress.com/2024/08/20/the-carbon-emissions-of-writing-and-illustrating-are-lower-for-ai-than-for-humans-report-february-2024/>

u/mobileJay77 • 1 points • 2025-03-11 20:28 UTC

I guess industrial use is way larger. The big players constantly have to train new models to stay ahead. Right now, someone is pouring venture capital through a cluster of GPUs hoping to become the next big thing.

u/05032-MendicantBias • 3 points • 2025-03-11 12:44 UTC

Training foundation models is expensive, but eventually VCs will get tired of throwing billions at the problem for no revenue.

As for inference, it's much cheaper to do generation than do it manually. If you consider a pic like this

[https://preview.redd.it/acpn8zci32oe1.jpeg?](https://preview.redd.it/acpn8zci32oe1.jpeg?width=2048&format=pjpg&auto=webp&s=dc91768e74ed716a4e7730fee3c04fd11e93f393)

[width=2048&format=pjpg&auto=webp&s=dc91768e74ed716a4e7730fee3c04fd11e93f393](https://preview.redd.it/acpn8zci32oe1.jpeg?width=2048&format=pjpg&auto=webp&s=dc91768e74ed716a4e7730fee3c04fd11e93f393)

It can take a few hours of photoshop, but it just needs a minute of GPU time, so it's going to enormously reduce energy use, not increase it.

u/InvestigatorNo8432 • 1 points • 2025-03-11 16:43 UTC

Jevons paradox shows that as energy efficiency increases, more people will be using it and therefore paradoxically increasing the overall energy used

u/CrungeBork • 1 points • 2025-03-11 15:46 UTC

This claim assumes that the number of images generated will remain the same and not increase in response to the higher throughput that's now possible.

The assumption is in direct disagreement with what we've seen in the field of computing and in American businesses.

u/ImYoric • 1 points • 2025-03-11 13:22 UTC

>It can take a few hours of photoshop, but it just needs a minute of GPU time, so it's going to enormously reduce energy use, not increase it.

That is true (and also how I justify to myself the use of Flux & co), but this near-immediacy of results means that we are considerably increasing the number of images (and musics, and videos, let's not forget them) that we produce. So, on an industrial scale, I think that we're much worse off.

Also, does this take into account the energy cost of building, shipping, installing, discarding these GPUs?

u/mervenca • 0 points • 2025-03-11 12:48 UTC

Yeah I get that in "time is money" principle, ai is way more efficient. But I meant just in terms of energy and water use.

A few hours on a low wattage laptop doesn't waste as much as a prompt in data centre? Or does it?

u/sothatsit • 1 points • 2025-03-12 04:48 UTC

\> A few hours on a low wattage laptop doesn't waste as much as a prompt in data centre? Or does it?

30W laptop for 3 hours: 324,000 Joules of energy

350W server for 15.5 seconds: 5425 Joules of energy (average time to generate a Flux image is 15.5 seconds)

So, a server generating a Flux image would use 1.6% of the energy of running a laptop for 3 hours. Even if you consider laptops that use less energy, generating flux images costs very little energy. The same is true for prompting an LLM. The servers just don't have to spend as much time processing your prompt as you might think (and often it may be processing multiple requests in parallel as well).

u/05032-MendicantBias • 2 points • 2025-03-11 12:53 UTC

I wouldn't call it wasting.

Netflix has enormous compute to stream their content. Datacenters use around 2% of the world power, which is a lot, but they also provide the internet, and that is providing incalculable value.

Running GPT and Copilot does use lots of energy, and water is used to cool those datacenters. There is a significant environmental impact.

The same apply to the internet as a whole.

I guess the key point is if that energy and cooling are spent wisely or not. I reckon yes, GenANI assist is very good at what it does, and can save energy compared to not using GenANI assist.

u/oruga_AI • 3 points • 2025-03-11 12:22 UTC

Nah, it uses less energy than a couple of episodes on netflix

u/mervenca • 1 points • 2025-03-11 12:44 UTC

Any proof of that?

u/StevenSamAI • 1 points • 2025-03-12 11:19 UTC

This sub is weird...

I felt the need to upvote this comment as it was on 0.

Is it a particularly good comment, no.

However OP is questioning things that are being put out as facts, and when someone then states something else as a fact with no explanation, OP gets downvoted for wanting some substance to that claim??

u/sothatsit • 1 points • 2025-03-12 02:09 UTC

Here's an article that talks about the water usage of ChatGPT:

<https://andymasley.substack.com/p/individual-ai-use-is-not-bad-for>

It argues that the energy usage of ChatGPT is *dwarfed* by the energy usage used for video streaming. That's not to say that the energy/water usage is not a problem, but rather that video streaming plays a much bigger role than AI. And that both are dwarfed by agriculture like cattle farming.

It bases the estimates of water usage on the energy consumption of ChatGPT and video streaming.

u/deadblankspacehole • 1 points • 2025-03-11 12:18 UTC

Complaining about environmental use triggers the need to "other" another group and it has good moral justification (if you don't think about it too much) so it's exciting to finger point about it while simultaneously using a slave labour device

u/AutoModerator • 1 points • 2025-03-11 12:13 UTC

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- * AI is going to take our jobs - its been asked a lot!
- * Discussion regarding positives and negatives about AI are allowed and encouraged. Just be respectful.
- * Please provide links to back up your arguments.
- * No stupid questions, unless its about AI being the beast who brings the end-times. It's not.

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